

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____ 10. _____ 11. _____ 12. _____

Function Families

Example (Name the parent & give the domain):

$f(x) = 2^x$ is the **exponential** parent: it passes through $(0, 1)$, never touches the x -axis, and its domain is all real numbers.

Directions: Name the parent family and state the domain.

1. $f(x) = x^2$

7. $f(x) = 2^x$

2. $f(x) = x^3$

8. $f(x) = \sqrt[3]{x}$

3. $f(x) = \sqrt{x}$

9. $f(x) = \frac{1}{x^2}$

4. $f(x) = \frac{1}{x}$

10. $f(x) = 5$

5. $f(x) = |x|$

11. $f(x) = x^4$

6. $f(x) = x$

Challenge (same sheet):

- 12.** $f(x) = \log_2 x$ — name the family, give the domain, and the one point every log parent passes through.

Answer Key — fully worked

Procedure: Match the rule to a known family; Recall the family's basic shape and a key point; State the domain — watch square roots and denominators.

1. Quadratic; domain all reals $(-\infty, \infty)$
2. Cubic; domain all reals
3. Square root; domain $x \geq 0$
4. Reciprocal (rational); domain $x \neq 0$
5. Absolute value; domain all reals
6. Linear (identity); domain all reals
7. Exponential; domain all reals
8. Cube root; domain all reals
9. Rational; domain $x \neq 0$
10. Constant; domain all reals
11. Quartic (even power); domain all reals
12. Logarithmic; domain $x > 0$; passes through $(1, 0)$